

Problem A. Kuriyama Mirai's Stones

Time Limit 2000 ms

Mem Limit 262144 kB

Input File stdin

Output File stdout

Kuriyama Mirai has killed many monsters and got many (namely n) stones. She numbers the stones from 1 to n . The cost of the i -th stone is v_i . Kuriyama Mirai wants to know something about these stones so she will ask you two kinds of questions:

1. She will tell you two numbers, l and r ($1 \leq l \leq r \leq n$), and you should tell her $\sum_{i=l}^r v_i$.
2. Let u_i be the cost of the i -th cheapest stone (the cost that will be on the i -th place if we arrange all the stone costs in non-decreasing order). This time she will tell you two numbers, l and r ($1 \leq l \leq r \leq n$), and you should tell her $\sum_{i=l}^r u_i$.

For every question you should give the correct answer, or Kuriyama Mirai will say "fuyukai desu" and then become unhappy.

Input

The first line contains an integer n ($1 \leq n \leq 10^5$). The second line contains n integers: $v_1, v_2, \dots, v_n$ ($1 \leq v_i \leq 10^9$) — costs of the stones.

The third line contains an integer m ($1 \leq m \leq 10^5$) — the number of Kuriyama Mirai's questions. Then follow m lines, each line contains three integers *type*, l and r ($1 \leq l \leq r \leq n$; $1 \leq \text{type} \leq 2$), describing a question. If *type* equal to 1 , then you should output the answer for the first question, else you should output the answer for the second one.

Output

Print m lines. Each line must contain an integer — the answer to Kuriyama Mirai's question. Print the answers to the questions in the order of input.

Examples

Input	Output
6 6 4 2 7 2 7 3 2 3 6 1 3 4 1 1 6	24 9 28

Input	Output
4 5 5 2 3 10 1 2 4 2 1 4 1 1 1 2 1 4 2 1 2 1 1 1 1 3 3 1 1 3 1 4 4 1 2 2	10 15 5 15 5 5 2 12 3 5

Note

Please note that the answers to the questions may overflow 32-bit integer type.

Problem B. Fence

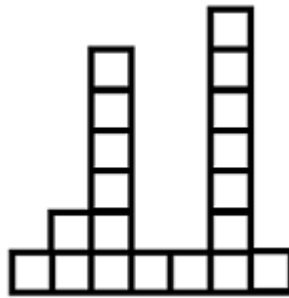
Time Limit 1000 ms

Mem Limit 262144 kB

Input File stdin

Output File stdout

There is a fence in front of Polycarpus's home. The fence consists of n planks of the same width which go one after another from left to right. The height of the i -th plank is h_i meters, distinct planks can have distinct heights.



Fence for $n = 7$ and $h = [1, 2, 6, 1, 1, 7, 1]$

Polycarpus has bought a posh piano and is thinking about how to get it into the house. In order to carry out his plan, he needs to take exactly k consecutive planks from the fence. Higher planks are harder to tear off the fence, so Polycarpus wants to find such k consecutive planks that the sum of their heights is minimal possible.

Write the program that finds the indexes of k consecutive planks with minimal total height. Pay attention, the fence is not around Polycarpus's home, it is in front of home (in other words, the fence isn't cyclic).

Input

The first line of the input contains integers n and k ($1 \leq n \leq 1.5 \cdot 10^5$, $1 \leq k \leq n$) — the number of planks in the fence and the width of the hole for the piano. The second line contains the sequence of integers $h_1, h_2, \dots, h_n$ ($1 \leq h_i \leq 100$), where h_i is the height of the i -th plank of the fence.

Output

Print such integer j that the sum of the heights of planks $j, j + 1, \dots, j + k - 1$ is the minimum possible. If there are multiple such j 's, print any of them.

Examples

Input	Output
7 3 1 2 6 1 1 7 1	3

Note

In the sample, your task is to find three consecutive planks with the minimum sum of heights. In the given case three planks with indexes 3, 4 and 5 have the required attribute, their total height is 8.

Problem C. Binary Deque

Time Limit 2000 ms

Mem Limit 262144 kB

Slavic has an array of length n consisting only of zeroes and ones. In one operation, he removes either the first or the last element of the array.

What is the minimum number of operations Slavic has to perform such that the total sum of the array is equal to s after performing all the operations? In case the sum s can't be obtained after any amount of operations, you should output -1 .

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The first line of each test case contains two integers n and s ($1 \leq n, s \leq 2 \cdot 10^5$) — the length of the array and the needed sum of elements.

The second line of each test case contains n integers a_i ($0 \leq a_i \leq 1$) — the elements of the array.

It is guaranteed that the sum of n over all test cases doesn't exceed $2 \cdot 10^5$.

Output

For each test case, output a single integer — the minimum amount of operations required to have the total sum of the array equal to s , or -1 if obtaining an array with sum s isn't possible.

Examples

Input	Output
7	0
3 1	1
1 0 0	3
3 1	2
1 1 0	2
9 3	7
0 1 0 1 1 1 0 0 1	-1
6 4	
1 1 1 1 1 1	
5 1	
0 0 1 1 0	
16 2	
1 1 0 0 1 0 0 1 1 0 0 0 0 0 1 1	
6 3	
1 0 1 0 0 0	

Note

In the first test case, the sum of the whole array is 1 from the beginning, so we don't have to make any operations.

In the second test case, the sum of the array is 2 and we want it to be equal to 1, so we should remove the first element. The array turns into $[1, 0]$, which has a sum equal to 1.

In the third test case, the sum of the array is 5 and we need it to be 3. We can obtain such a sum by removing the first two elements and the last element, doing a total of three operations. The array turns into $[0, 1, 1, 1, 0, 0]$, which has a sum equal to 3.

Problem D. Range Xor Queries

Time Limit 1000 ms

Mem Limit 524288 kB

Given an array of n integers, your task is to process q queries of the form: what is the xor sum of values in range $[a, b]$?

Input

The first input line has two integers n and q : the number of values and queries.

The second line has n integers $x_1, x_2, \dots, x_n$: the array values.

Finally, there are q lines describing the queries. Each line has two integers a and b : what is the xor sum of values in range $[a, b]$?

Output

Print the result of each query.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$
- $1 \leq a \leq b \leq n$

Example

Input	Output
8 4	3
3 2 4 5 1 1 5 3	0
2 4	6
5 6	4
1 8	
3 3	

Problem E. Balance

Time Limit 2000 ms

Problem Statement

We have N weights indexed 1 to N . The mass of the weight indexed i is W_i .

We will divide these weights into two groups: the weights with indices not greater than T , and those with indices greater than T , for some integer $1 \leq T < N$. Let S_1 be the sum of the masses of the weights in the former group, and S_2 be the sum of the masses of the weights in the latter group.

Consider all possible such divisions and find the minimum possible absolute difference of S_1 and S_2 .

Constraints

- $2 \leq N \leq 100$
- $1 \leq W_i \leq 100$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

```
N
W1 W2 ... WN-1 WN
```

Output

Print the minimum possible absolute difference of S_1 and S_2 .

Sample 1

Input	Output
3 1 2 3	0

If $T = 2$, $S_1 = 1 + 2 = 3$ and $S_2 = 3$, with the absolute difference of 0.

Sample 2

Input	Output
4 1 3 1 1	2

If $T = 2$, $S_1 = 1 + 3 = 4$ and $S_2 = 1 + 1 = 2$, with the absolute difference of 2. We cannot have a smaller absolute difference.

Sample 3

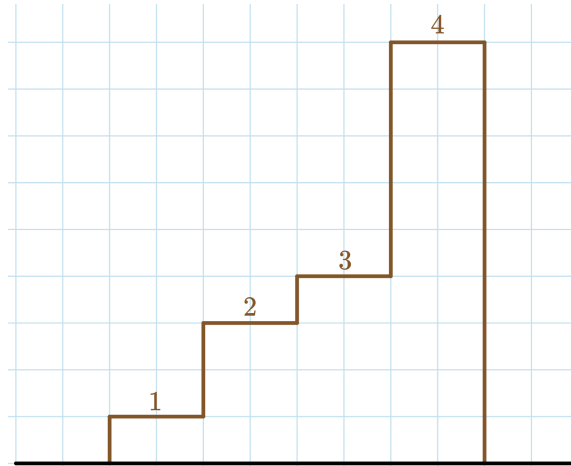
Input	Output
8 27 23 76 2 3 5 62 52	2

Problem F. Scuza

Time Limit 3000 ms

Mem Limit 262144 kB

Timur has a stairway with n steps. The i -th step is a_i meters higher than its predecessor. The first step is a_1 meters higher than the ground, and the ground starts at 0 meters.



The stairs for the first test case.

Timur has q questions, each denoted by an integer $k_1, \dots, k_q$. For each question k_i , you have to print the maximum possible height Timur can achieve by climbing the steps if his legs are of length k_i . Timur can only climb the j -th step if his legs are of length at least a_j . In other words, $k_i \geq a_j$ for each step j climbed.

Note that you should answer each question independently.

Input

The first line contains a single integer t ($1 \leq t \leq 100$) — the number of test cases.

The first line of each test case contains two integers n, q ($1 \leq n, q \leq 2 \cdot 10^5$) — the number of steps and the number of questions, respectively.

The second line of each test case contains n integers ($1 \leq a_i \leq 10^9$) — the height of the steps.

The third line of each test case contains q integers ($0 \leq k_i \leq 10^9$) — the numbers for each question.

It is guaranteed that the sum of n does not exceed $2 \cdot 10^5$, and the sum of q does not exceed $2 \cdot 10^5$.

Output

For each test case, output a single line containing q integers, the answer for each question.

Please note, that the answer for some questions won't fit into 32-bit integer type, so you should use at least 64-bit integer type in your programming language (like `long long` for C++).

Examples

Input	Output
3 4 5 1 2 1 5 1 2 4 9 10 2 2 1 1 0 1 3 1 1000000000 1000000000 1000000000 1000000000	1 4 4 9 9 0 2 3000000000

Note

Consider the first test case, pictured in the statement.

- If Timur's legs have length 1, then he can only climb stair 1, so the highest he can reach is 1 meter.
- If Timur's legs have length 2 or 4, then he can only climb stairs 1, 2, and 3, so the highest he can reach is $1 + 2 + 1 = 4$ meters.
- If Timur's legs have length 9 or 10, then he can climb the whole staircase, so the highest he can reach is $1 + 2 + 1 + 5 = 9$ meters.

In the first question of the second test case, Timur has no legs, so he cannot go up even a single step. : (

Problem G. Max $A_i + B_j$

Time Limit 2000 ms

Problem Statement

You are given two integer sequences A and B , each of length N . Choose integers i, j ($1 \leq i, j \leq N$) to maximize the value of $A_i + B_j$.

Constraints

- $1 \leq N \leq 5 \times 10^5$
- $|A_i| \leq 10^9$ ($i = 1, 2, \dots, N$)
- $|B_j| \leq 10^9$ ($j = 1, 2, \dots, N$)
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
N
A1 A2 ... AN
B1 B2 ... BN
```

Output

Print the maximum possible value of $A_i + B_j$.

Sample 1

Input	Output
2 -1 5 3 -7	8

For $(i, j) = (1, 1), (1, 2), (2, 1), (2, 2)$, the values of $A_i + B_j$ are 2, -8 , 8, -2 respectively, and $(i, j) = (2, 1)$ achieves the maximum value 8.

Sample 2

Input	Output
6 15 12 3 -13 -1 -19 7 17 -13 -10 18 4	33

Problem H. Black and White Stripe

Time Limit 2000 ms

Mem Limit 262144 kB

You have a stripe of checkered paper of length n . Each cell is either white or black.

What is the minimum number of cells that must be recolored from white to black in order to have a segment of k consecutive black cells on the stripe?

If the input data is such that a segment of k consecutive black cells already exists, then print 0.

Input

The first line contains an integer t ($1 \leq t \leq 10^4$) — the number of test cases.

Next, descriptions of t test cases follow.

The first line of the input contains two integers n and k ($1 \leq k \leq n \leq 2 \cdot 10^5$). The second line consists of the letters 'W' (white) and 'B' (black). The line length is n .

It is guaranteed that the sum of values n does not exceed $2 \cdot 10^5$.

Output

For each of t test cases print an integer — the minimum number of cells that need to be repainted from white to black in order to have a segment of k consecutive black cells.

Examples

Input	Output
4	1
5 3	2
BBWBW	0
5 5	1
BBWBW	
5 1	
BBWBW	
1 1	
W	

Note

In the first test case, $s = \text{"BBWBW"}$ and $k = 3$. It is enough to recolor s_3 and get $s = \text{"BBBBW"}$. This string contains a segment of length $k = 3$ consisting of the letters 'B'.

In the second test case of the example $s = \text{"BBWBW"}$ and $k = 5$. It is enough to recolor s_3 and s_5 and get $s = \text{"BBBBB"}$. This string contains a segment of length $k = 5$ consisting of the letters 'B'.

In the third test case of the example $s = \text{"BBWBW"}$ and $k = 1$. The string s already contains a segment of length $k = 1$ consisting of the letters 'B'.

Problem I. Static Range Sum Queries

Time Limit 1000 ms

Mem Limit 524288 kB

Given an array of n integers, your task is to process q queries of the form: what is the sum of values in range $[a, b]$?

Input

The first input line has two integers n and q : the number of values and queries.

The second line has n integers $x_1, x_2, \dots, x_n$: the array values.

Finally, there are q lines describing the queries. Each line has two integers a and b : what is the sum of values in range $[a, b]$?

Output

Print the result of each query.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$
- $1 \leq a \leq b \leq n$

Example

Input	Output
8 4 3 2 4 5 1 1 5 3 2 4 5 6 1 8 3 3	11 2 24 4

Problem J. Good Subarrays

Time Limit 2000 ms

Mem Limit 262144 kB

You are given an array $a_1, a_2, \dots, a_n$ consisting of integers from 0 to 9. A subarray $a_l, a_{l+1}, a_{l+2}, \dots, a_{r-1}, a_r$ is good if the sum of elements of this subarray is equal to the length of this subarray ($\sum_{i=l}^r a_i = r - l + 1$).

For example, if $a = [1, 2, 0]$, then there are 3 good subarrays: $a_{1..1} = [1]$, $a_{2..3} = [2, 0]$ and $a_{1..3} = [1, 2, 0]$.

Calculate the number of good subarrays of the array a .

Input

The first line contains one integer t ($1 \leq t \leq 1000$) — the number of test cases.

The first line of each test case contains one integer n ($1 \leq n \leq 10^5$) — the length of the array a .

The second line of each test case contains a string consisting of n decimal digits, where the i -th digit is equal to the value of a_i .

It is guaranteed that the sum of n over all test cases does not exceed 10^5 .

Output

For each test case print one integer — the number of good subarrays of the array a .

Examples

Input	Output
3	3
3	6
120	1
5	
11011	
6	
600005	

Note

The first test case is considered in the statement.

In the second test case, there are 6 good subarrays: $a_{1..1}$, $a_{2..2}$, $a_{1..2}$, $a_{4..4}$, $a_{5..5}$ and $a_{4..5}$.

In the third test case there is only one good subarray: $a_{2..6}$.